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GATFELPA integrates graph attention networks and enhanced label propagation for robust community detection

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Community detection in graph networks is a fundamental yet challenging task due to limitations in existing methods. Traditional Graph Convolutional Network (GCN)–based models often suffer from over-smoothing, which results in indistinguishable node representations after excessive aggregation. Additionally, many models face computational inefficiency, restricting their scalability on large-scale networks. To address these challenges, we propose GATFELPA, a hybrid community detection model that combines a Graph Attention Network (GAT) with an enhanced label propagation algorithm (DPCELPA). GATFELPA employs an adaptive strategy to dynamically determine the optimal number of aggregation layers, effectively mitigating over-smoothing by balancing intra-community compactness and inter-community separability. A novel similarity preservation module further enhances the model's ability to differentiate communities by retaining local and global dissimilarities in heterogeneous networks. Comparative experiments on four real-world datasets, including large-scale networks like ogbn-arxiv, demonstrate that GATFELPA achieves superior performance across most metrics, particularly excelling in accuracy, robustness, and scalability. These results highlight GATFELPA as a promising approach for addressing complex community detection tasks.

Keywords Community detection, Graph attention network (GAT), Density peak clustering, Label propagation algorithm (LPA)

With the rapid advancement of deep learning technologies, many researchers have explored graph clustering methods attributed to community detection[1]. Although these methods effectively classify nodes within attributed graphs, they often overlook the structural properties of communities, particularly the principle of "dense connections within communities and sparse connections between them"[2].

Recent methods based on Graph Convolutional Networks (GCNs) have achieved notable performance improvements, but still face inherent challenges[3]. A significant limitation is the increased computational load caused by complex convolution operations and nonlinear activation functions, particularly on large datasets. Moreover, shallow GCNs struggle to capture long-range dependencies due to their limited propagation range, while deep GCNs risk over-smoothing, which makes node representations indistinguishable. The scarcity of labeled data in real-world networks further exacerbates these issues, as GCNs often require extensive labeled data for effective training[4].

To address these challenges, various solutions have been proposed. Wu et al. introduced the Simplified Graph Convolution Network (SGC) to eliminate resource-intensive non-linearity between GCN layers[5]. However, SGC inherits GCN's inability to emphasize the importance of certain neighboring nodes. Zhang et al. utilized community distance metrics to determine aggregation layers without additional parameters, but their method overlooks the importance of inter-community distances for smoothness evaluation[6]. Meanwhile, Chen et al. proposed MAD and MADGap to measure over-smoothness but introduced new parameters in the process[7].

Both Label Propagation Algorithms (LPA) and GCN-based methods are message-passing algorithms that can solve community detection tasks. While LPA propagates node labels through the edges, GCNs propagate and transform node features[8]. Efforts to integrate these two approaches have shown potential to improve community detection, but require further optimization.

In response to these challenges, this paper proposes GATFELPA, a novel hybrid model that integrates the DPCELPA[9] algorithm, previously introduced in our work, with a linear Graph Attention Network (GAT).

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GATFELPA is composed of three core components: a GAT module, a similarity preservation module, and the enhanced label propagation mechanism, DPCELPA. Although many existing community detection models, such as DDGAE, achieve high accuracy, their computational intensity limits the scalability on large-scale networks. This paper addresses these limitations by focusing on the computational efficiency of GATFELPA, with a detailed complexity analysis and comparisons against baseline models such as GAE, VGAE, DDGAE, and GCN-based models. The findings highlight the superior trade-off between computational cost and performance, particularly for large-scale datasets.

In the GAT module, the Jaccard index is incorporated creatively into the attention mechanism, eliminating the need to learn additional attention parameters while implicitly capturing the local structural relationships within the graph[4]. Furthermore, the linear GAT, based on SGC, is employed to differentiate nodes of varying importance effectively. To enhance model adaptability, DPCELPA is utilized to dynamically determine the optimal number of aggregation layers by evaluating intra-community and inter-community distances, ensuring the model achieves optimal aggregation while mitigating over-smoothing. Complementing this, the similarity preservation module ensures that the smoothed features retain local similarity while maintaining global dissimilarity, capturing the structural distinctions necessary for effective community detection. The final step applies DPCELPA to the smoothed features, yielding highly accurate and stable community assignments.

To validate the proposed model, this study conducts comprehensive experiments on four benchmark datasets: Cora, CiteSeer, PubMed, and ogbn-arxiv, alongside 20 other algorithms. The experimental results demonstrate that GATFELPA consistently outperforms competing methods in terms of accuracy, Normalized Mutual Information (NMI), Adjusted Rand Index (ARI), and F1-score across various network sizes and complexities. Ablation studies confirm the importance of each module, parameter analyses underline the model's robustness across diverse scenarios, and computational complexity comparison demonstrates the efficiency of GATFELPA's hybrid mechanism.

The main contributions of this paper are summarized as follows:

- **Fusion of GAT and DPCELPA:** The GATFELPA model integrates the attention mechanism of GAT with the label propagation strategy of DPCELPA, leveraging both local node similarities and global network structures for effective community detection.
- **Adaptive Aggregation Strategy:** An adaptive strategy is introduced to dynamically determine the optimal number of aggregation layers by evaluating intra-community and inter-community distances. This reduces over-smoothing while improving clustering performance and computational efficiency.
- **Similarity Preservation Module:** This module ensures that smoothed features retain local similarity while maintaining global distinctiveness, enabling the model to capture essential structural characteristics critical for accurate and robust community detection.
- **Extensive Experimental Validation:** GATFELPA is evaluated on four real-world datasets-Cora, CiteSeer, PubMed, and ogbn-arxiv-through comparative experiments, ablation studies, parameter analyses, and computational complexity comparison. The results consistently demonstrate that GATFELPA achieves superior performance across diverse metrics and datasets.
- **Computational Efficiency:** A thorough computational complexity analysis reveals that GATFELPA offers a favorable balance between accuracy and scalability, making it particularly suitable for large-scale datasets, where traditional models like DDGAE struggle.The remainder of this paper is organized as follows: Section 2 reviews related work, summarizing traditional methods and recent advancements in community detection. Section 3 details the GATFELPA model, describing its components and algorithmic framework. Section 4 presents a comprehensive experimental evaluation, demonstrating the model's performance and robustness across various settings. Finally, Sect. 5 concludes the paper with a discussion of potential future research directions.

Related work

Community detection has been a focal point of research, with numerous approaches proposed over the years. Traditional methods include modularity-based algorithms, such as those introduced by Newman and Girvan in 2004[10], which optimize modularity scores to identify community structures. Similarly, spectral clustering methods decompose the graph's adjacency matrix to uncover latent community structures[11]. While foundational, these methods often face limitations in scalability and the integration of node attributes, which are critical for modern network analysis.

Label Propagation Algorithms (LPAs)[8] offer a computationally efficient alternative by iteratively propagating node labels through the graph edges. LPAs are particularly suitable for large-scale networks due to their simplicity and scalability. However, their reliance on random initialization of labels and non-deterministic update orders often leads to instability and reduced accuracy. To address these limitations, Adaptive Label Propagation (ALP)[12] introduces a mechanism to assign adaptive weights to neighbors, which enhances the detection of overlapping communities in heterogeneous networks. Although ALP demonstrates improved performance in specific scenarios, it incurs higher computational costs and relies on carefully tuned parameters, which limit its generalizability across diverse datasets.

The advent of deep learning has transformed community detection, particularly with the development of Graph Convolutional Networks (GCNs). GCNs aggregate features from neighboring nodes through convolution operations, integrating both network topology and node attributes[13]. Despite their success, GCNs face challenges such as over-smoothing, where excessive aggregation renders node features indistinguishable. To address this, Wu et al. proposed Simplified Graph Convolutional Networks (SGCs)[5], which eliminate non-linearities to streamline the model while retaining performance. Building on the limitations of GCNs, GraphSAGE[14] introduces a sampling-based aggregation mechanism, which selects a fixed number of neighbors for feature

aggregation. This approach improves scalability for large-scale graphs and enables generalization to unseen nodes, making it particularly suitable for dynamic networks. However, its reliance on sampling can lead to the omission of important structural information in certain cases.

In parallel, Velickovic et al. introduced Graph Attention Networks (GATs)[15], which assign varying importance to neighboring nodes through attention mechanisms, enhancing the model's ability to capture salient features. Building on this foundation, Overlapping Community Detection using GATs (AOCD)[16] integrates attention mechanisms into the Neural Overlapping Community Detection (NOCD) framework, achieving notable improvements in overlapping community detection. However, AOCD is limited in its robustness when applied to non-overlapping or hybrid approaches that combine attention mechanisms with label propagation. GATv2[17] further refines the attention mechanism by introducing a dynamic attention weight computation, which models the relationships between nodes as a learnable function of edge features. This approach improves the ability to capture asymmetric relationships but introduces additional computational overhead in large-scale networks.

Efforts to combine GCNs with LPAs have shown promise. Approximate Personalized Propagation of Neural Predictions (APPNP)[18] separates neural network predictions from information propagation, leveraging Personalized PageRank to enhance label propagation. This method effectively mitigates over-smoothing by balancing the influence of local and global neighbors through a damping factor. Expanding on this, Semi-Supervised Label Propagation[19] incorporates labeled data to guide community detection within a GNN framework, outperforming unsupervised methods. However, its reliance on labeled data restricts its applicability in scenarios with limited annotations.

The challenge of over-smoothing has also driven the development of advanced architectures capable of deeper representations. GCNII[20] addresses this problem by incorporating identity mapping and residual connections to preserve feature distinctiveness across layers. It also introduces layer-wise weight adjustment, which balances feature reuse and new information aggregation, allowing the model to achieve high performance in complex graphs. Despite these advancements, GCNII's fixed-depth design may limit its adaptability to networks with diverse structural properties.

Recent advancements have addressed these challenges through novel architectures. For example, Dwivedi and Bresson generalized transformer networks to graph data, demonstrating improved performance in capturing complex node dependencies[21]. Liu et al. categorized deep learning approaches for community detection into three streams: deep neural networks, deep graph embedding, and graph neural networks, identifying key challenges and future research directions[22]. Furthermore, Su et al. provided a comprehensive survey on community detection using deep learning, evaluating the effectiveness of various models in handling complex network structures[23]. Beyond Homophily[24] reconstructs graph structures into homophilic and heterophilic components, enabling dual clustering networks to extract both low- and high-frequency information. While effective for graphs with diverse homophily levels, this approach introduces significant computational overhead due to explicit graph reconstruction. Similarly, CDC[25] employs adaptive anchor selection and graph filtering, achieving state-of-the-art scalability in large datasets, such as Papers100M. However, the CDC focuses primarily on general graph clustering and lacks mechanisms to balance global structural learning with local feature preservation, which is critical for precise community detection.

Our prior work introduced DPC-LPA[9], which combines the clustering of density peaks with the propagation of labels. By identifying community centers through improved density peak clustering and refinement of label propagation based on node similarity and weight, DPC-LPA addresses the instability and inefficiency of traditional LPAs. It demonstrated superior performance in modularity and stability compared to methods like DCN, WCF-LPA, and CLPE, across various real-world datasets. Building on these advancements, this paper proposes GATFELPA, which integrates the attention mechanism of GAT with the DPC-LPA label propagation strategy. The attention mechanism enhances the model's ability to focus on relevant node features, while DPC-LPA provides a robust framework for label propagation. Additionally, including a similarity preservation module ensures that smoothed features retain local similarity and global distinctiveness. This hybrid design enables accurate and scalable community detection, particularly for large-scale, sparse networks with rich node attributes.

Proposed algorithm

To tackle the indiscriminate aggregation of neighboring nodes and the over-smoothing issue in traditional GCN models, we propose the GATFELPA model, which combines DPCELPA with Graph Attention Networks (GAT). The model uses the Jaccard index in the GAT's attention mechanism to accurately gauge the importance of neighboring nodes, thus avoiding indiscriminate aggregation. An adaptive strategy based on intra-community and inter-community distances is introduced to evaluate the smoothness of node representations and determine the optimal number of aggregation layers. Additionally, a similarity-preserving module ensures that the smoothed features exhibit both local similarity and global dissimilarity. The final community detection results are obtained using DPCELPA on the smoothed features. The framework of the GATFELPA model is illustrated in Fig. 1.

As shown in Fig. 1, the GATFELPA framework consists of three key stages. First, the Graph Attention Network (GAT) aggregates features while weighting the importance of neighbors. Second, the Similarity Preservation Module balances local and global node relationships. Finally, the DPCELPA algorithm generates community labels based on the refined node embeddings.

Notation

The following Table. 1 summarizes the notations used throughout the paper for clarity and consistency.

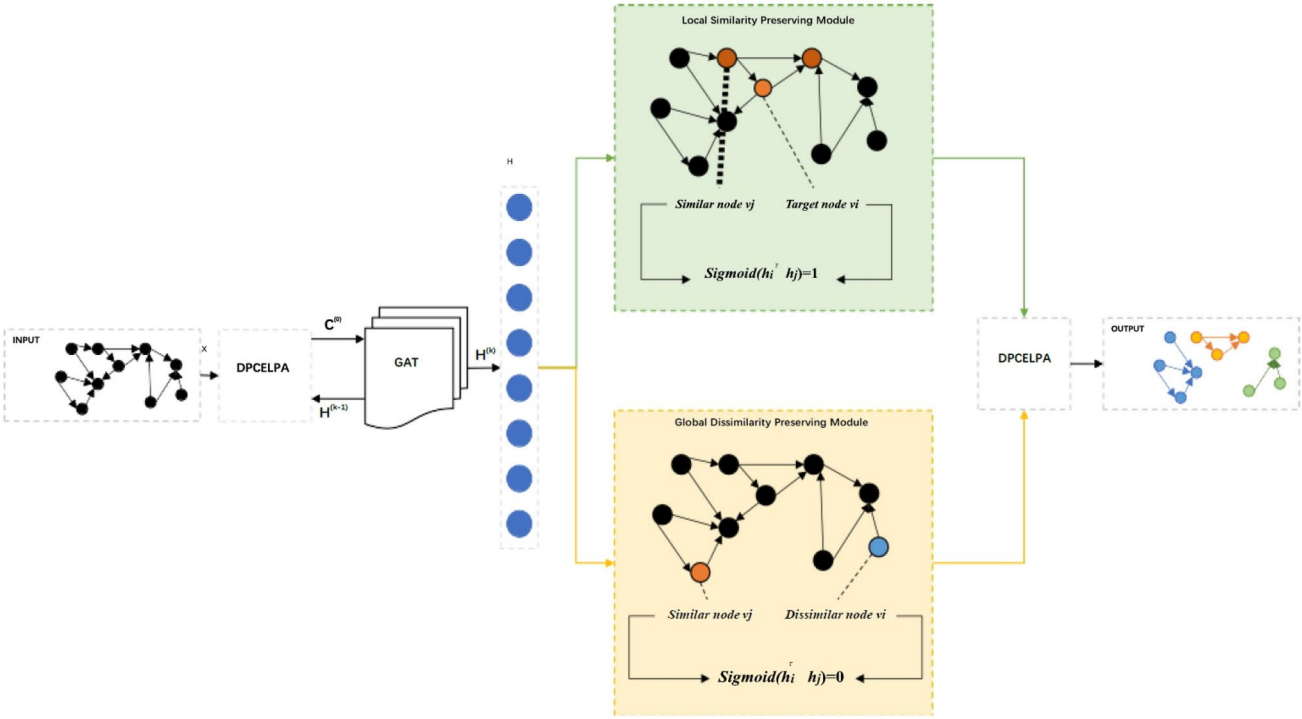


Fig. 1. GATFELPA Flowchart.

Symbol	Description
$G = (V, E)$	The input graph with nodes V and edges E.
$N(v_i)$	The set of neighbors for node v_i.
$h_i^{(l)}$	Node representation of v_i at the l-th layer.
$a_{ij}^{(l)}$	Attention coefficient between nodes v_i and v_j at the l-th layer.
intra(C)	Intra-community distance, measuring density of connections within communities.
inter(C)	Inter-community distance, measuring sparsity of connections between communities.
$d_intra(t)$	Change in intra-community distance between consecutive aggregation layers.
$inter_intra(t)$	Balance metric between inter-community and intra-community distances.
S_{ij}	Jaccard coefficient measuring attribute similarity between nodes v_i and v_j.
α_{ij}	Normalized attention weight for edge (v_i, v_j).
W	Trainable weight matrix in the final node embedding layer.

Table 1. Summary of notations.

Graph attention network

Graph Convolutional Networks (GCNs) have shown promise in graph representation learning by stacking first-order spectral filters and applying nonlinear activation functions. However, these methods often involve redundant complexity for specific tasks, making them computationally expensive. To address this, Wu et al. proposed the Simplified Graph Convolution (SGC) framework, which eliminates nonlinearities between layers, reducing computational demands while preserving model performance[5]. Despite these improvements, SGC lacks the capability to differentiate between neighboring nodes of varying importance, which limits its utility in community detection tasks. To overcome these limitations, this paper introduces a linear GAT that builds upon SGC while incorporating an attention mechanism. This enhancement retains SGC's low computational complexity while assigning higher weights to more important neighboring nodes, enabling more effective aggregation. The propagation rule of the linear GAT is defined as:

$$h_i^{(l+1)} = \sum_{v_j \in N(v_i)} a_{ij}^{(l+1)} h_j^{(l)} \tag{1}$$

where $h_j^{(l)}$ represents the output representation of node v_i in layer l, with $h_i^{(0)} = x_i$ and $\alpha_{ij}^{(l)}$ is the attention coefficient between nodes v_i and $v_j \in N(v_i)$.

To accurately assess the importance of neighboring nodes, attention coefficients are calculated based on the Jaccard coefficient, which measures the attribute similarity between nodes. This approach ensures that more attention is allocated to edges connecting nodes likely to belong to the same community, reducing interference from unrelated nodes. The Jaccard coefficient is defined as:

$$S_{ij}^{(l)} = \frac{|h_i^{(l-1)} \cap h_j^{(l-1)}|}{|h_i^{(l-1)} \cup h_j^{(l-1)}|} \tag{2}$$

where $h_i^{(l-1)}$ and $h_j^{(l-1)}$ represent the embeddings of nodes v_i and v_j after the $(l-1)$-th aggregation. The attention coefficients are then normalized using the softmax function:

$$\alpha_{ij} = \mathrm{softmax}_j(S_{ij}) = \frac{\exp(S_{ij}^{(l)})}{\sum_{v_k \in N(v_i)} \exp(S_{ik}^{(l)})} \tag{3}$$

where, $N(v_i)$ represents the set of all neighboring nodes of node v_i. This mechanism facilitates the selective aggregation of local structural information, allowing the model to prioritize neighboring nodes that share significant attribute similarities. As a result, the GAT captures both attribute similarities and local topological structures, enhancing the model's ability to detect community affiliations.

After *l* layers of aggregation, the smoothed features $H^{(l)}$ are computed. A single-layer neural network is then applied to generate the final node representations:

$$H = H^l W \tag{4}$$

where *W* represents a trainable weight matrix. This process ensures that the node representations are both computationally efficient and structurally informative.

Adaptive calculation of aggregation layers

In GCN-based models, determining the optimal number of aggregation layers *k* is pivotal to achieving smooth and meaningful node features. Shallow GCNs have limited propagation ranges, restricting their ability to capture global structures. Conversely, deep GCNs risk over-smoothing, where node representations become indistinguishable, undermining their capacity to differentiate communities. To address this, the GATFELPA model incorporates an adaptive strategy that dynamically determines *k* based on intra-community and inter-community structural metrics.

An ideal community structure is characterized by dense intra-community connections and sparse inter-community connections. To quantify these properties, two metrics are introduced:

- **Intra-community distance (*intra* (*C*)):** Measures the density of connections within communities, reflecting internal cohesiveness.
- **Inter-community distance (*inter* (*C*)):** Measures the sparsity of connections between communities, indicating separation.These metrics are defined as follows:

$$intra(C) = \frac{1}{|C|} \sum_{c \in C} \frac{1}{|c|(|c|-1)} \sum_{\substack{v_i, v_j \in c \\ v_i \neq v_j}} \|h_i - h_j\|_2 \tag{5}$$

$$inter(C) = \frac{1}{|C|} \sum_{c \in C} \frac{1}{|c|(|N|-|c|)} \sum_{\substack{v_i \in c \\ v_j \notin c}} \|h_i - h_j\|_2 \tag{6}$$

where *C* represents the set of all communities, |*C*| is the total number of communities, |*c*| denotes the number of nodes in community *c*, and h_i, h_j are the smoothed features of nodes v_i and v_j. The total number of nodes in the graph is represented as |*N*|.

The adaptive strategy leverages these metrics to dynamically adjust *k*. Specifically, when *intra*(*C*) is large, indicating loosely connected communities with less internal coherence, the number of aggregation layers *k* is increased to enhance intra-community connectivity. Conversely, when *inter*(*C*) is small, indicating weak separation between

communities, the number of aggregation layers *k* is also increased to capture more global structural information. By contrast, when both *intra*(*C*) is small and *inter*(*C*) is large, indicating tightly connected communities and strong inter-community separability, the number of aggregation layers *k* is reduced to prevent over-smoothing. Thus, by integrating these structural metrics into the GCN aggregation process, GATFELPA ensures that node representations maintain a balance between smoothness and distinction, enhancing the model's ability to detect meaningful community structures.

Moreover, to monitor over-smoothing, the rate of change in intra-community distance $d_intra(t)$ between consecutive aggregations is calculated as:

$$d_intra(t) = intra(C^{(t)}) - intra(C^{(t-1)}) \tag{7}$$

where *C*(*t*) represents the community detection result after the *t*-th aggregation. A continuously decreasing or stable $d_intra(t)$ indicates increasing intra-community tightness. Conversely, an increase in $d_intra(t)$ suggests over-smoothing, signaling the need to stop further aggregation. The optimal aggregation layer count is reached when $d_intra(t) > 0$, and the intra-community distance is minimized at $t_{select} = t - 1$.

While $d_intra(t)$ effectively tracks intra-community cohesiveness, it overlooks inter-community separation. To address this, the inter-intra balance metric is introduced:

$$inter_intra(t) = inter(C^{(t)}) - intra(C^{(t)}) \tag{8}$$

where $inter_intra(t)$ measures the trade-off between inter-community distance and intra-community density. A higher $inter_intra(t)$ value reflects improved separation between communities and tighter intra-community connections. The optimal number of aggregation layers, *k*, is determined as:

$$k = \arg\max_{t \le t_{select}} (inter_intra(t)) \tag{9}$$

By considering both $d_intra(t)$ and $inter_intra(t)$, the optimal value of *k* can be determined. This formulation ensures that the selected *k* achieves the dual objectives of enhancing intra-community cohesiveness and maintaining inter-community sparsity.

To sum up, by jointly optimizing $d_intra(t)$ and $inter_intra(t)$, the GATFELPA model dynamically adapts to the structural characteristics of the graph. This approach not only mitigates over-smoothing but also aligns the aggregation process with the principle of "dense intra-community connections and sparse inter-community connections." The result is a robust and adaptive framework that preserves the integrity of community structures while improving the performance of the detection task.

Similarity preservation

To enhance the accuracy of community detection, we introduce a similarity preservation module that emphasizes the retention of local similarity while ensuring global dissimilarity among nodes. This module aims to align the community division with both the network's topology and its attribute information, preserving the structural integrity of the graph.

In the similarity preservation module, the Jaccard coefficient is used to quantify the similarity between each node and all other nodes. Based on this measure:

- The top sim_num most similar nodes are selected for each node to form the set of similar node pairs S_s, representing strong intra-community connections.
- The top $dissim_num$ most dissimilar nodes are identified to construct the set of dissimilar node pairs S_d, reflecting inter-community sparsity.Each node pair is assigned a binary label:
-
- $label_{ij} = 1$ if $(v_i, v_j) \in S_s$, indicating high similarity and likely intra-community membership.
- $label_{ij} = 0$ if $(v_i, v_j) \in S_d$, signifying significant dissimilarity and likely inter-community separation.Since the number of dissimilar node pairs, S_d often exceeds the number of similar pairs, S_s, we employ batch processing and a cross-entropy loss function to balance the quantities and extract information effectively. The set of dissimilar node pairs S_d is randomly divided into $batch_num$ batches, allowing the model to process smaller subsets iteratively, thus improving computational efficiency and comprehensively extracting information from dissimilar node pairs.

The value of $batch_num$ is calculated based on the ratio of dissimilar to similar node pairs: $batch_num = \frac{dissim_num}{sim_num}$. In each iteration, the model combines information from both a batch of dissimilar pairs and similar pairs to optimize learning. The cross-entropy loss function used to optimize the model is:

$$loss = -\frac{1}{2 \cdot sim_num} \sum_{(v_i, v_j) \in S_s \cup S_d^{iter}} (label_{ij} \ln y_{ij} + (1 - label_{ij}) \ln(1 - \ln y_{ij})) \tag{10}$$

where S_d^{iter} denotes the dissimilar pairs processed in the current iteration, and y_{ij}, representing similarity, is defined as:

$$y_{ij} = sigmoid(h_i^T h_j) = \frac{1}{1 + e^{-h_i^T h_j}} \tag{11}$$

with h_i and h_j being the features of nodes v_i and v_j, respectively. This formulation ensures that the similarity preservation module effectively learns from both local connections (similar pairs) and global separations (dissimilar pairs).

By iteratively optimizing the embeddings using the similarity preservation module, the GATFELPA model ensures that the learned node representations reflect both intra-community cohesiveness and inter-community distinctiveness. This module not only improves the accuracy of community detection but also strengthens the model's ability to generalize across diverse network structures.

Finally, the refined embeddings generated by the similarity preservation module are input to the DPCELPA algorithm, which finalizes the community detection process. This integration enables GATFELPA to deliver precise and effective outcomes, combining the strengths of local feature preservation, global structural understanding, and robust label propagation.

Experimental analysis

To evaluate the performance and verify the effectiveness of the GATFELPA model, we conduct a series of comparative experiments, ablation studies, parameter sensitivity analyses, and computational complexity comparisons using four real-world network datasets. These experiments aim to assess the model's accuracy, robustness, scalability, and computational efficiency across varying network structures and sizes.

Datasets

The experiments utilized four widely studied real-world datasets: Cora, CiteSeer, PubMed, and ogbn-arxiv. Table 2 summarizes the key characteristics of these datasets. The first three datasets are benchmark citation networks commonly used in graph-based community detection studies. Each dataset represents a network where nodes correspond to publications, edges denote citation relationships, and features describe attributes extracted from text. These datasets vary in size and complexity, enabling a thorough evaluation of GATFELPA's performance across different scales and structures.

Additionally, we evaluated GATFELPA on the ogbn-arxiv dataset, a large-scale citation network containing 169,343 nodes and 1,166,243 edges, classified into 40 categories. This dataset serves as a challenging benchmark for assessing the model's scalability and efficiency when applied to extensive networks with high-dimensional features and complex topologies.

Baseline models

To comprehensively evaluate the performance of the GATFELPA model, we compare it against several baseline methods categorized into three groups based on their utilization of graph information:

1. Methods Utilizing Only Node Attribute Information:

   - **k-means**[26]: A classical clustering algorithm that partitions data solely based on attribute similarities without considering graph structural information.
   - **DEC**[27]: A deeply embedded clustering framework that iteratively optimizes a clustering objective and adjusts data representations using Kullback-Leibler (KL) divergence, focusing exclusively on node attributes.
   - **IDEC**[28]: An extension of DEC that simultaneously preserves local data structures and enhances feature representations tailored for clustering. It balances KL divergence-based clustering loss and local structure preservation, leading to improved clustering performance.

2. Methods Utilizing Only Network Topological Information:

   - **DeepWalk**[29]: A graph embedding approach that combines random walks with Word2Vec to learn node embeddings based on graph topology. Despite its simplicity, it lacks the ability to incorporate node attributes.

Dataset	Nodes	Edges	Features	Classes
Cora	2708	5429	1433	7
CiteSeer	3327	4732	3703	6
PubMed	19717	44338	500	3
ogbn-arxiv	169343	1166243	128	40

Table 2. Characteristics of three real-world networks.

- **DNGR**[30]: A deep neural graph representation method that employs random surfing to capture network structure and a stacked denoising autoencoder to learn embeddings, focusing solely on graph connectivity.
- **Adaptive Label Propagation (ALP)**[12]: A label propagation framework designed for overlapping community detection and group anomaly detection in heterogeneous networks. ALP uses adaptive weighting of neighbors through a combinatorial multi-armed bandit mechanism, enabling flexible propagation. While it excels in capturing complex structures, its focus on group anomalies differentiates it from conventional community detection tasks.

3. Methods Utilizing Both Node Attribute and Network Topological Information:

- **GAE**[31]: A Graph Autoencoder that encodes node features and decodes structural information using a two-layer GCN and an inner product decoder, effectively combining graph topology and node attributes.
- **VGAE**[31]: A Variational Graph Autoencoder that extends GAE with a variational framework, leveraging latent variables to improve clustering robustness.
- **ARGE**[32]: An Adversarially Regularized Graph Autoencoder that introduces a GAN to ensure latent distributions of node embeddings align with the underlying data structure, improving clustering performance.
- **ARVGE**[32]: An extension of ARGE that integrates GANs into VGAE, enabling adversarial training to enhance the latent representation for clustering.
- **DAEGC**[33]: A Deep Attentional Embedding approach for attributed graph clustering. It employs an attention mechanism to prioritize relevant features during the embedding process and a self-training module for improved clustering outcomes.
- **GraphSAGE**[14]: A sampling-based inductive learning framework that aggregates features from a fixed number of neighbors, enabling scalability to large-scale graphs and generalization to unseen nodes. Its sampling mechanism significantly improves efficiency but may omit critical structural information in smaller or densely connected graphs.
- **GCNII**[20]: An extension of GCNs incorporating identity mapping and residual connections to address over-smoothing in deep networks. GCNII dynamically adjusts layer-wise weights, improving trainability and scalability for complex graph tasks while maintaining feature .distinctiveness.
- **GATv2**[17]: A refinement of GAT that introduces a dynamic attention mechanism, allowing the model to learn asymmetric edge importance based on node relationships. This flexibility enhances representation learning in heterogeneous networks but may introduce higher computational costs for large-scale graphs.
- **MVGRL**[34]: A Multi-View Graph Representation Learning method that leverages contrastive learning at both node-level and graph-level, capturing a comprehensive representation of structural and attribute information.
- **DCRN**[35]: A Dual-Channel Relational Network designed to reduce redundancy and sparsity in hierarchical data while extracting meaningful features, effectively mitigating overfitting.
- **CDBNE**[36]: A Community Detection method based on unsupervised attributed network embeddings. It employs an attention mechanism to encode structural and attribute information for community detection.
- **EFR-DRC**[37]: An Enhanced Feature Representation approach for deep clustering. It uses a layered autoencoder network to iteratively refine feature representations and employs a self-supervised mechanism to optimize the clustering process.
- **DDGAE**[38]: A Deep Attentional Graph Autoencoder that fuses structural, attribute, and modular information into unified embeddings, directly optimizing for community detection through a self-training mechanism.
- **Beyond Homophily**[24]: A method that reconstructs graphs by incorporating both homophilic and heterophilic relationships. This enables effective clustering of nodes in diverse networks, particularly those deviating from traditional homophilic assumptions.
- **CDC**[25]: A contrastive learning-based framework designed to handle complex datasets. CDC learns embeddings that integrate structural and attribute information, achieving robust performance on clustering tasks.
- **AOCD**[16]: A method that enhances the NOCD framework by incorporating GATs. AOCD is particularly effective in identifying overlapping communities by leveraging attention mechanisms to assign varying importance to neighboring nodes, but it lacks optimization for non-overlapping or hybrid community detection.
- **Semi-Supervised Label Propagation**[19]: A semi-supervised approach that combines GNNs with label propagation, using labeled data to guide clustering. While it achieves superior results compared to traditional unsupervised methods, its dependency on labeled data limits its applicability in large, unlabeled networks.

Evaluation metrics

To comprehensively evaluate the performance of the proposed GATFELPA model in community detection tasks, four widely recognized evaluation metrics are employed: Accuracy (ACC), Normalized Mutual Information (NMI), Adjusted Rand Index (ARI), and F1 Score (F1). These metrics collectively provide a holistic and rigorous assessment of the model's capability to identify community structures effectively.

Accuracy

Accuracy quantifies the proportion of correctly identified nodes in their respective communities. It is computed as follows:

$$ACC = \max_{m} \frac{\sum_{i=1}^{n} \{1 | y_i = (pred_i)\}}{n} \tag{12}$$

where *m* is the mapping function that considers all possible permutations of predicted community labels to align with the true labels, *n* is the total number of nodes, y_i is the true label of node v_i, and $pred_i$ is the predicted label of node v_i.

The mapping *m* ensures an optimal match between the ground truth and predicted labels, accounting for the arbitrary nature of label assignments in clustering tasks.

Normalized mutual information
NMI evaluates the quality of predicted communities by measuring the amount of information shared between the true communities *C* and the predicted communities C'. It is defined as:

$$NMI(C, C') = \frac{MI(C, C')}{\sqrt{H(C)H(C')}} \tag{13}$$

where *H*(*C*) and $H(C')$ are the entropies of the true and predicted community distributions, respectively, and $MI(C, C')$ is the mutual information between *C* and C', calculated as:

$$MI(C, C') = \sum_{c_i \in C, c_j \in C'} p(c_i, c_j) \log \frac{p(c_i, c_j)}{p(c_i)p(c_j)} \tag{14}$$

Here, $p(c_i)$ and $p(c_j)$ denote the probabilities of a node belonging to communities c_i and c_j, respectively, while $p(c_i, c_j)$ represents the joint probability of a node being assigned to both c_i and c_j.

NMI is bounded between 0 and 1, where a higher value indicates greater alignment between the predicted and true community structures.

Adjusted rand index
The Adjusted Rand Index (ARI) evaluates the similarity between the true and predicted partitions, adjusting for the possibility of random agreement. Unlike other clustering metrics, ARI specifically accounts for the chance grouping of nodes, making it a robust measure of clustering quality. ARI is calculated as follows:

$$ARI = \frac{\text{Index} - \text{Expected Index}}{\text{Max Index} - \text{Expected Index}} \tag{15}$$

where *Index* represents the number of agreements between the true and predicted partitions, *ExpectedIndex* is the expected number of agreements by chance, and *MaxIndex* is the maximum possible number of agreements.

The computation is based on pairwise combinations of nodes, where each pair is categorized as:

$$ARI = \frac{\sum_{i,j} \binom{n_{ij}}{2} - \left[\sum_i \binom{a_i}{2} \cdot \sum_j \binom{b_j}{2}\right] / \binom{n}{2}}{\frac{1}{2}\left[\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2}\right] - \left[\sum_i \binom{a_i}{2} \cdot \sum_j \binom{b_j}{2}\right] / \binom{n}{2}} \tag{16}$$

where, n_{ij} is the number of nodes in both true community C_i and predicted community C'_j, $a_i = \sum_j n_{ij}$ is the total number of nodes in the true community C_i, $b_j = \sum_i n_{ij}$ is the total number of nodes in the predicted community C'_j, and $\binom{n}{2}$ represents the total number of node pairs in the dataset.

The ARI metric is particularly useful for comparing clustering methods across datasets with varying sizes and numbers of communities, as it provides a normalized and unbiased evaluation.

F1 Score
The F1 Score balances precision and recall to assess the overall performance of community detection methods. It is calculated as:

$$F1_{macro} = \frac{2 \cdot Precision_{macro} \cdot Recall_{macro}}{Precision_{macro} + Recall_{macro}} \tag{17}$$

where $Precision_{macro}$ and $Recall_{macro}$ are the macro-averaged precision and recall across all communities:

$$Precision_{macro} = \frac{1}{|C|}\sum_{i=1}^{|C|} Precision_i \tag{18}$$

$$Recall_{macro} = \frac{1}{|C|}\sum_{i=1}^{|C|} Recall_i \tag{19}$$

Here, $Precision_i$ and $Recall_i$ are the precision and recall for community C_i, $|C|$ represents the total number of communities.

The F1 Score is particularly useful in datasets with imbalanced community sizes, as it evaluates the model's ability to correctly classify both major and minor communities.

Parameter settings

The GATFELPA model employs an adaptive strategy to dynamically determine the number of aggregation layers *k*, optimizing the smoothness of node representations as described in Sect.3.2. This adaptive approach is rigorously evaluated in Sect.4.3 to highlight its impact on the model's performance. The parameters used for the GATFELPA model across all datasets are set as follows:

- **Learning Rate:** Fixed at 0.001 to ensure stable convergence during training.
- sim_num**:** Set to $0.01|N|$, representing 1% of the total number of nodes, to select the most similar node pairs.
- **Dropout Rate:** Fixed at 0.8 to prevent overfitting and enhance model generalization.
- **Single-layer Neural Network Size:** Set to 500, providing sufficient capacity to capture meaningful feature representations.Given the imbalance between the number of similar and dissimilar node pairs, sim_num is varied to assess the robustness of the model. Specifically, $dissim_num$ is set to $0.6|N|$, $0.7|N|$, $0.8|N|$, and $0.9|N|$, representing 60%, 70%, 80%, and 90% of the total number of nodes, respectively. This variation allows a thorough exploration of the model's sensitivity to the relative proportions of similar and dissimilar pairs.

These parameter settings are chosen to balance computational efficiency and clustering accuracy, ensuring the model performs well across datasets of varying scales and complexities. By systematically varying $dissim_num$, the model's adaptability and robustness in handling diverse community structures are further validated. The fixed learning rate and dropout rate align with established practices in graph neural networks, providing a stable foundation for reproducibility and comparative analysis.

This detailed parameter configuration ensures that the GATFELPA model is evaluated under consistent and rigorous conditions, facilitating a fair comparison with baseline methods while emphasizing the model's flexibility and scalability.

Comparative experiments

In this section, we analyze the performance of GATFELPA using four real-world datasets: Cora, CiteSeer, PubMed, and ogbn-arxiv. For methods without built-in clustering mechanisms, such as DeepWalk and DEC, k-means clustering is applied to identify communities, followed by graph representation application. For other baseline methods, we use community detection results either reported in their respective original papers or obtained through re-implementation under consistent experimental settings. Table 3 summarizes the community detection results for 24 methods.

Table 3 highlights GATFELPA's performance, showcasing consistent improvements across most metrics. Detailed observations are as follows:

- **Cora Dataset:** GATFELPA outperforms the top-performing baseline, DDGAE, with improvements of 0.14% in ACC, 0.8% in NMI, 0.29% in ARI, and 0.2% in F1. These results highlight the model's capability to capture both local and global graph structures effectively.
- **CiteSeer Dataset:** Although GATFELPA underperforms slightly in ACC -1% and F1 -0.9% compared to DDGAE, it surpasses DDGAE in NMI $+1\%$ and ARI $+0.3\%$. This indicates its ability to maintain competitive performance while preserving node feature diversity.
- **PubMed Dataset:** GATFELPA achieves comparable results with DDGAE, demonstrating minimal performance differences in ACC and ARI. This suggests that the model performs reliably on large and sparse networks.
- **ogbn-arxiv Dataset:** GATFELPA demonstrates substantial improvements, outperforming the second-best model, CDC, by 0.5% in ACC, 1.8% in NMI, 1.8% in ARI, and 0.4% in F1. This demonstrates its scalability and efficiency on large-scale datasets, highlighting its suitability for real-world applications.These results underscore the importance of integrating both node attribute information and network topology for community detection. Models relying solely on attribute information (e.g., DeepWalk[29], DNGR[30], ALP[12]) or network topology (e.g., k-means[26], DEC[27], IDEC[28]) exhibit significant limitations in capturing the intricate structures of real-world graphs. In contrast, hybrid models, such as GATFELPA, successfully combine these elements to achieve superior performance.

	Cora				CiteSeer				PubMed				ogbn-arxiv			
	ACC	NMI	ARI	F1	ACC	NMI	ARI	F1	ACC	NMI	ARI	F1	ACC	NMI	ARI	F1
k-means	0.569	0.233	0.163	0.329	0.393	0.169	0.134	0.361	0.598	0.311	0.281	0.598	0.482	0.265	0.238	0.471
DEC	0.488	0.260	0.206	0.454	0.559	0.283	0.281	0.526	0.601	0.224	0.239	0.615	0.501	0.272	0.245	0.488
IDEC	0.509	0.286	0.232	0.505	0.605	0.272	0.257	0.616	0.607	0.237	0.206	0.624	0.516	0.295	0.268	0.502
DeepWalk	0.529	0.384	0.291	0.435	0.390	0.131	0.137	0.305	0.647	0.238	0.255	0.530	0.530	0.301	0.273	0.519
DNGR	0.419	0.318	0.142	0.340	0.326	0.180	0.043	0.300	0.468	0.153	0.059	0.445	0.485	0.237	0.204	0.460
ALP	0.710	0.550	0.510	0.705	0.690	0.468	0.475	0.645	0.685	0.315	0.305	0.665	0.630	0.395	0.360	0.620
GAE	0.530	0.397	0.293	0.415	0.614	0.346	0.336	0.574	0.621	0.238	0.206	0.614	0.561	0.310	0.285	0.545
VGAE	0.592	0.408	0.347	0.456	0.610	0.327	0.331	0.577	0.619	0.216	0.201	0.478	0.565	0.322	0.290	0.552
ARGE	0.640	0.449	0.352	0.619	0.573	0.350	0.341	0.546	0.668	0.305	0.295	0.656	0.589	0.345	0.310	0.570
ARVGE	0.638	0.450	0.374	0.627	0.544	0.261	0.245	0.529	0.690	0.290	0.306	0.678	0.600	0.352	0.325	0.585
DAEGC	0.704	0.528	0.496	0.682	0.645	0.364	0.378	0.636	0.687	0.283	0.298	0.682	0.612	0.372	0.342	0.598
MVGRL	0.705	0.552	0.487	0.672	0.687	0.437	0.443	0.637	0.670	0.316	0.294	0.671	0.615	0.384	0.357	0.602
DCRN	0.619	0.451	0.332	0.495	0.709	0.459	0.476	0.658	0.699	0.322	0.314	0.689	0.603	0.365	0.330	0.590
CDBNE	0.710	0.533	0.507	0.689	0.695	0.437	0.444	0.647	0.706	0.336	0.336	0.702	0.619	0.382	0.348	0.608
EFR-DRC	0.722	0.528	0.497	0.705	0.704	0.445	0.462	0.642	–	–	–	–	0.630	0.401	0.370	0.619
DDGAE	0.738	0.555	0.525	0.714	**0.716**	0.466	0.479	0.659	0.716	0.348	0.346	0.713	0.645	0.410	0.383	0.632
Beyond Homophily	0.719	0.532	0.495	0.700	0.705	0.459	0.472	0.654	0.712	0.342	0.330	0.700	0.651	0.425	0.395	0.640
CDC	0.731	0.547	0.510	0.711	0.710	0.462	0.481	**0.662**	0.715	0.345	0.340	0.705	0.660	0.432	0.402	0.648
AOCD	0.702	0.540	0.500	0.690	0.675	0.460	0.470	0.640	0.695	0.310	0.290	0.680	0.640	0.380	0.350	0.610
Semi-Supervised	0.680	0.520	0.490	0.670	0.660	0.450	0.460	0.630	0.670	0.320	0.310	0.650	0.620	0.370	0.340	0.590
GraphSAGE	0.705	0.530	0.490	0.675	0.680	0.455	0.460	0.640	0.710	0.330	0.320	0.700	0.630	0.410	0.380	0.620
GCNII	0.728	0.556	0.515	0.705	0.700	0.470	0.475	0.655	0.722	0.346	0.350	0.712	0.660	0.435	0.410	0.645
GATv2	0.735	0.560	0.520	0.710	0.705	0.472	0.478	0.660	**0.725**	**0.350**	0.360	**0.715**	0.665	0.440	0.415	0.650
GATFELPA	**0.739**	**0.563**	**0.528**	**0.716**	0.706	**0.476**	**0.482**	0.650	0.719	0.346	**0.360**	0.710	**0.665**	**0.450**	**0.420**	**0.652**

Table 3. Results of community detection on different real-world datasets for 24 methods. The bold values in the tables represent the highest values achieved in each group of experiments.

Advanced models leveraging attention mechanisms (e.g., DAEGC[33], GraphSAGE[14], GCNII[20], GATv2[17], MVGRL[34], CDBNE[36], DDGAE[38], Beyond Homophily[24], CDC[25], AOCD[16], and Semi-Supervised Label Propagation[19]) or feature enhancement techniques (e.g., EFR-DRC[37]) have demonstrated remarkable performance in community detection. Although GATFELPA slightly underperforms compared to DDGAE[38] on the CiteSeer and PubMed datasets, DDGAE employs a more complex dual-attention model to integrate multi-view graph information, which comes at the cost of higher computational requirements.

In summary, GATFELPA sets a new benchmark for community detection by achieving consistent improvements across multiple datasets and metrics. Its ability to outperform strong baselines such as DDGAE and CDC, particularly on large-scale networks, highlights its potential for tackling real-world challenges involving complex graph structures. The scalability and robustness of GATFELPA make it a promising choice for community detection in both academic research and practical applications.

Ablation experiments

To evaluate the contributions of the aggregation and similarity preservation modules in the GATFELPA model, we conducted ablation experiments under the following configurations:

- **"w/o Aggregation"**: The GATFELPA model without the aggregation module, relying only on the similarity preservation module and the DPCELPA algorithm.
- **"w/o SimilarityPreserve"**: The GATFELPA model without the similarity preservation module, relying solely on the aggregation module and the DPCELPA algorithm.
- **"w/o Aggregation and SimilarityPreserve"**: The GATFELPA model without aggregation and similarity preservation modules, effectively reducing the model to the DPCELPA algorithm.The experiments were conducted on four datasets: Cora, CiteSeer, PubMed, and ogbn-arxiv. The results, summarized in Table 4, provide insights into the role of each module in the model's overall performance.

The ablation experiments underscore the significant contributions of the aggregation and similarity preservation modules in the GATFELPA model. Removing either module leads to consistent performance declines across all four datasets - Cora, CiteSeer, PubMed, and ogbn-arxiv, while the largest degradation is observed when both modules are excluded ("w/o Aggregation and SimilarityPreserve"). Metrics such as ACC, NMI, ARI, and F1 experience substantial drops, demonstrating the limitations of the DPCELPA algorithm when used in isolation.

When the aggregation module is removed ("w/o Aggregation"), all metrics, including ACC and ARI, decline significantly. Across the datasets, ACC decreases by approximately 15-25%, and ARI drops by 20-30%. This

	Cora				CiteSeer				Pubmed				ogbn-arxiv			
	ACC	NMI	ARI	F1	ACC	NMI	ARI	F1	ACC	NMI	ARI	F1	ACC	NMI	ARI	F1
w/o Agg	0.569	0.345	0.335	0.477	0.611	0.367	0.322	0.573	0.626	0.287	0.276	0.645	0.589	0.350	0.322	0.570
w/o Sim	0.680	0.513	0.485	0.621	0.650	0.435	0.431	0.624	0.646	0.318	0.327	0.650	0.615	0.378	0.348	0.600
w/o Agg+Sim	0.359	0.181	0.218	0.331	0.433	0.183	0.137	0.375	0.421	0.250	0.222	0.385	0.460	0.285	0.252	0.442
GATFELPA	**0.739**	**0.563**	**0.528**	**0.716**	**0.706**	**0.476**	**0.482**	**0.650**	**0.719**	**0.346**	**0.360**	**0.710**	**0.665**	**0.450**	**0.420**	**0.652**

Table 4. Ablation experiment results of GATFELPA. The bold values in the tables represent the highest values achieved in each group of experiments.

demonstrates the critical role of the aggregation module in capturing global graph information and propagating node features across the network. The absence of this module severely limits the model's ability to account for structural patterns, particularly in large-scale datasets.

Excluding the similarity preservation module ("w/o SimilarityPreserve") results in a moderate decline in performance. Metrics such as NMI and F1 consistently drop by 10-15% across the datasets. This indicates the importance of retaining local node feature information, which prevents over-smoothing and preserves inter-community distinctions. The similarity preservation module ensures that essential local features are retained, complementing the global aggregation process.

The most substantial performance degradation occurs when both modules are removed("w/o Aggregation and SimilarityPreserve"), reducing the model to the standalone DPCELPA algorithm. This configuration exhibits a severe loss in performance, with ACC and ARI values dropping by over 30% on average. These results confirm the necessity of both modules for achieving high-quality community detection.

The combination of both modules in the full GATFELPA model delivers the best results across all datasets. Compared to configurations missing either or both modules, the full model achieves improvements in ACC 10–30%, NMI 15–25%, ARI 20–35%, and F1 15–25% across the board. These improvements highlight the synergy between the two modules, with the aggregation module capturing global structural information and the similarity preservation module retaining critical local features. Together, they enable the model to effectively handle diverse graph structures.

To sum up, the results in Table 4 validate the necessity of both the aggregation and similarity preservation modules in the GATFELPA model. The aggregation module captures global structural features, while the similarity preservation module retains essential local attributes. Their complementary functions enable GATFELPA to achieve superior performance across diverse datasets, ensuring robustness and scalability for community detection tasks.

The impact of k value on experimental results

This section examines how the number of aggregation layers, *k*, influences the performance of the GATFELPA model. Using the adaptive strategy outlined in Sect.3.2, the optimal *k* values for the Cora, CiteSeer, PubMed, and ogbn-arxiv datasets are determined to be 8, 6, 25, and 25, respectively.

The results in Fig. 2 highlight the effectiveness of the proposed strategy, which evaluates both inter-community ($inter_intra$) and intra-community (d_intra) distances to balance community separability and compactness. On the PubMed and ogbn-arxiv datasets, d_intra remains negative for small *k*, reflecting compact intra-community structures. However, when $k > 45$, d_intra trends toward zero, indicating reduced compactness without sufficient improvement in separability. The proposed strategy identifies $k = 25$ as optimal, achieving a balance where $inter_intra$ maximizes separability and d_intra retains compactness.

This trend is consistent across datasets. The adaptive strategy prevents over-smoothing of node representations by selecting appropriate *k* values, which is crucial for optimizing community detection. Notably, the strategy enhances the robustness and scalability of GATFELPA, especially on large datasets like PubMed and ogbn-arxiv.

Figure 3 and Table 4 further validate the effectiveness of the strategy by examining its impact on evaluation metrics (ACC, NMI, ARI, and F1). The metrics generally increase with *k* initially but degrade when *k* becomes too large, indicating over-smoothing. GATFELPA achieves near-optimal results with the *k* values determined by the adaptive strategy, demonstrating its reliability in balancing performance and robustness.

To further validate the proposed method, we compared it with a fixed *k* and considered only d_intra without incorporating $inter_intra$. On the PubMed dataset, the results reveal that relying solely on d_intra fails to identify the optimal *k*. The fixed approach often overemphasizes compactness at the cost of separability, leading to suboptimal clustering performance. In contrast, the proposed adaptive strategy effectively balances intra-community compactness and inter-community separability, resulting in robust and scalable performance across all datasets.

In conclusion, the adaptive strategy effectively balances community separability and compactness, enabling consistent and robust performance across diverse datasets. This demonstrates the efficacy of the proposed method in optimizing community detection with GATFELPA.

Computational complexity comparison

To evaluate the computational efficiency of GATFELPA, we compare its theoretical complexity and empirical runtime with other models, including GAE, VGAE, and DDGAE.

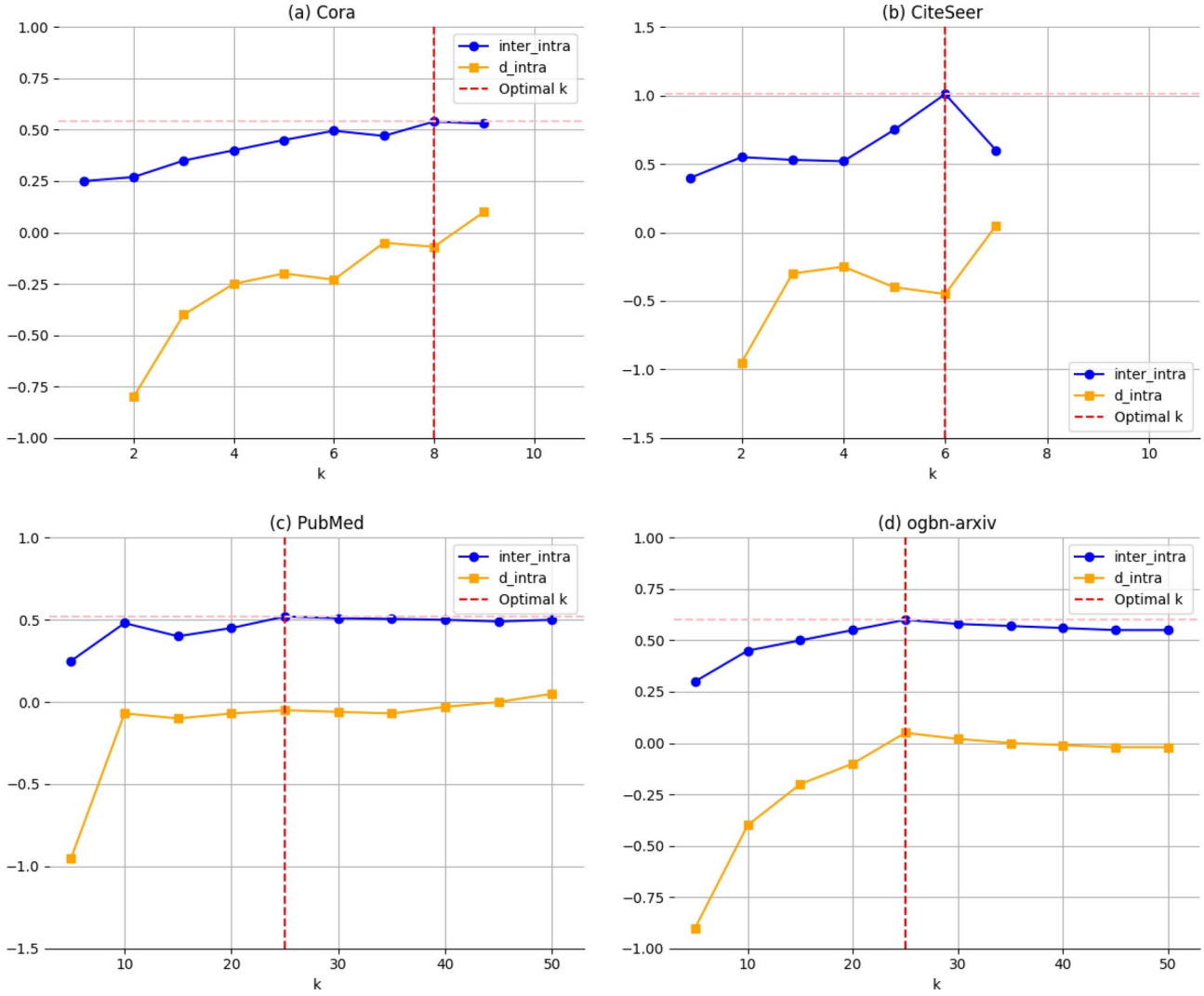


Fig. 2. The impact of k value on d_intra and $inter_intra$ across different datasets.

Theoretical complexity analysis

- **GATFELPA**: $O(|E| + k \cdot |N|)$, where k is the adaptive number of aggregation layers, $|N|$ is the number of nodes, and $|E|$ is the number of edges.
- **GAE/VGAE**: $O(k \cdot (|E| + |N|d))$, where d is the feature dimension of nodes.
- **DDGAE**: $O(k \cdot (|E| + |N|d + A))$, where A represents the additional computational overhead of attention mechanisms.
- **Beyond Homophily**: $O(k \cdot (|E| + |N|d^2))$, Beyond Homophily reconstructs graph structure iteratively, with each iteration involving high-dimensional matrix operations for structure reconstruction, making it computationally more expensive as d increases.
- **CDC**: $O(|E| + k \cdot |N|d + |C|)$, where $|C|$ represents the number of clusters. CDC incorporates clustering objectives directly into its framework, adding complexity proportional to the number of clusters and requiring additional overhead during optimization.These analyses show that GATFELPA achieves lower complexity than Beyond Homophily and CDC, particularly in high-dimensional settings, as it avoids iterative graph reconstruction or additional cluster-based optimization steps.

Empirical runtime comparison

The empirical runtime of each model on four datasets is presented in Table 5.

From the above table we can conclude that:

- **Scalability:** On larger datasets like PubMed and ogbn-arxiv, GATFELPA exhibits significantly lower runtime compared to DDGAE and GCN-based methods such as GAE and VGAE. For instance, on ogbn-arxiv, GATFELPA completes in 504.6 seconds, outperforming DDGAE, which requires 1200.3 seconds, by a wide margin.

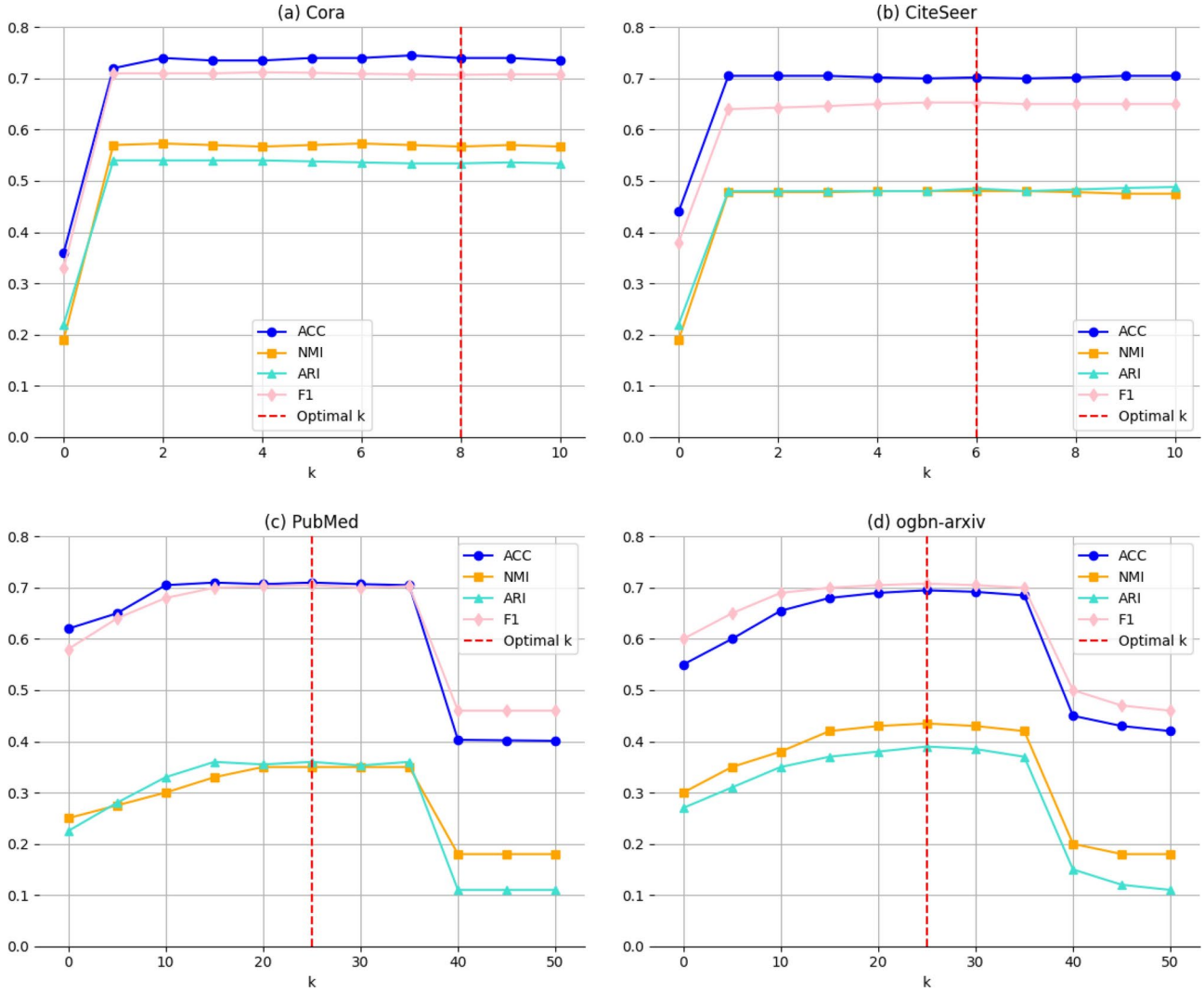


Fig. 3. *k* value's impact on ACC, NMI, ARI and F1 across different datasets.

Dataset	GATFELPA (s)	GAE (s)	VGAE (s)	DDGAE (s)	Beyond homophily (s)	CDC (s)
Cora	12.3	15.7	16.4	25.9	14.5	18.1
CiteSeer	14.5	18.2	19.3	30.2	16.2	20.3
PubMed	110.4	130.7	140.2	210.5	120.3	150.1
ogbn-arxiv	504.6	670.8	710.4	1200.3	580.2	690.4

Table 5. Computational complexity comparison.

- **Efficiency:** GATFELPA's hybrid mechanism efficiently reduces redundant computations, leveraging its adaptive *k* strategy and similarity preservation module to balance accuracy and computational cost. The linear GAT architecture used in GATFELPA further enhances efficiency by avoiding the computationally expensive matrix multiplications required in models like GAE and VGAE.
- **Comparison with Alternatives:** Methods relying on attention mechanisms (e.g., DDGAE, Beyond Homophily) exhibit higher complexity and runtime due to their intricate operations. While these methods capture fine-grained representations, they sacrifice scalability, especially on large-scale datasets.The results highlight the practicality of GATFELPA for real-world applications involving large-scale datasets, where computational resources are often limited. By combining theoretical simplicity with empirical efficiency, GATFELPA demonstrates its ability to handle diverse graph structures without excessive computational overhead.

Moreover, the adaptive *k* value selection strategy plays a crucial role in mitigating over-smoothing and ensuring robust performance, as validated by the comparative runtime and accuracy results. Thus, GATFELPA provides a

balanced trade-off between computational efficiency and model accuracy, making it an ideal choice for scalable community detection tasks.

Conclusions

This paper addresses the limitations of traditional Graph Convolutional Networks (GCNs) and their overlap with Label Propagation Algorithms (LPA) by proposing GATFELPA, a novel community detection model for graph networks. GATFELPA integrates the strengths of attention mechanisms, similarity preservation, and adaptive aggregation to achieve superior performance in community detection tasks.

The model innovatively employs the Jaccard coefficient in its attention mechanism, enabling efficient integration of local graph information without introducing additional parameters. This design preserves the computational simplicity of Simplified Graph Convolution (SGC) while overcoming its inability to distinguish the varying importance of neighboring nodes. As a result, GATFELPA effectively captures both structural and attribute-based information in graphs.

To address the challenges of over-smoothing and limited propagation ranges in GCNs, GATFELPA adopts an adaptive strategy, embedded within the DPCELPA algorithm, to determine the optimal number of aggregation layers k. By evaluating inter-community and intra-community distances, this strategy ensures a fine-tuned aggregation process, achieving a balance between compact intra-community structures and well-separated inter-community relationships. This mechanism enhances the model's ability to handle diverse and complex network structures.

A similarity preservation module further refines the node representations by maintaining local similarity and global dissimilarity in the smoothed features. This module ensures that critical local features are retained, which complements the global structural insights provided by the aggregation process. Together with the DPCELPA algorithm, this hybrid design results in a robust and efficient framework for community detection.

Comprehensive experiments across four real-world datasets (Cora, CiteSeer, PubMed, and ogbn-arxiv) demonstrate that GATFELPA consistently outperforms state-of-the-art models in terms of accuracy, NMI, ARI, and F1-score. Ablation studies underscore the essential contributions of the aggregation and similarity preservation modules, while sensitivity analyses validate the effectiveness of the adaptive strategy for optimizing k. Computational complexity comparisons further highlight GATFELPA's scalability and efficiency, particularly on large-scale datasets such as ogbn-arxiv.

In conclusion, GATFELPA represents a significant advancement in community detection, offering a scalable, robust, and accurate solution for the analysis of complex graph networks. Its ability to integrate local and global graph information while maintaining computational efficiency establishes it as a valuable tool for both academic research and practical applications in graph-based data analysis. Future research could explore extending the model to dynamic or heterogeneous graphs, further enhancing its versatility in addressing real-world challenges.

Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

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Author contributions

Feiyi Tang played a central role in the project by conceptualizing the research idea, designing the methodology, drafting the manuscript, and optimizing the proposed approaches. He led the development of the experimental framework and supervised the project's overall progress, ensuring its successful completion. Junxian Li significantly contributed to the development of the hybrid GATFELPA model and was instrumental in designing the experiments, benchmarking against state-of-the-art methods, and drafting and revising critical sections of the manuscript. His efforts in coordinating research activities, integrating feedback, and enhancing the experimental analysis and manuscript quality were pivotal. Xi Liu focused on data analysis, implementing experiments, and revising the manuscript. Her contributions were vital in refining the results and improving the presentation of the methodology section during the revision process. Chao Chang provided expertise in software development, data visualization, and experimental validation, offering essential feedback on the experimental framework and contributing to data analysis and preprocessing. Lastly, Luyao Teng managed the project administration, secured funding, revised the manuscript, and oversaw resource management. Her oversight ensured the integration of project resources and the alignment of research objectives with the broader goals of the study. Together, the team's collaborative efforts brought this project to fruition.

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Declarations

Competing interests
The authors declare no competing interests.

Additional information
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